OEAS 705/805 Advanced Environmental Data Science

Fall 2026

October 22, 2026

Dr. Jessie Turner, jturners@odu.edu

Class times: OCNPS 320, T/Th, 11:00 - 12:15

Office hours: After class T/Th,

And: OCNPS 411, M, 1:00 - 3:00



Schedule

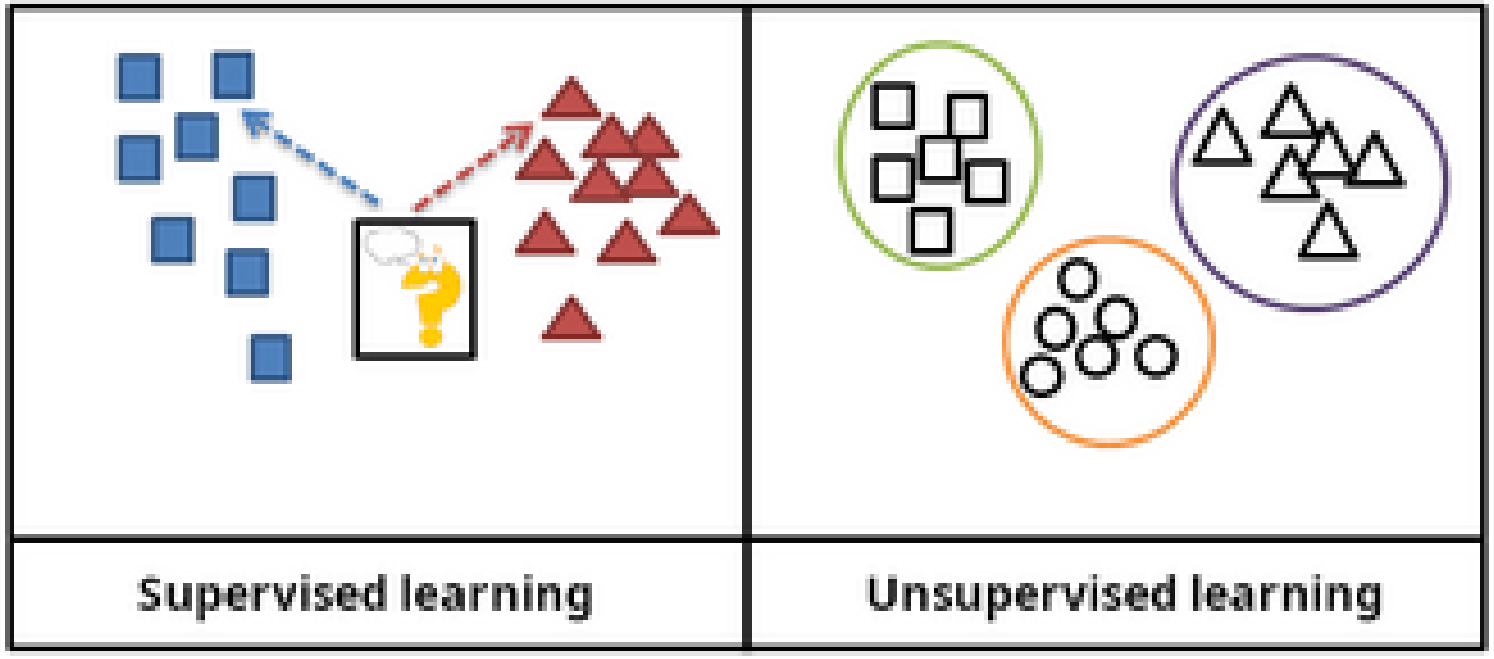
- HW 4 will be posted soon - Due Monday November 2nd.
- No class Tuesday November 3rd for Election Day.
- Projects complete November 23rd for Peer Review
- Peer Reviews Due by November 30th (graded for completion only)

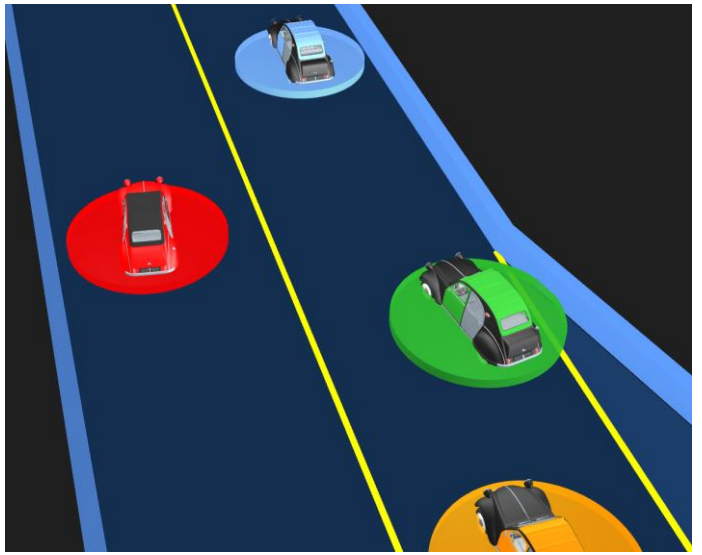
Machine Learning

1. Supervised Learning

2. Unsupervised Learning

3. Reinforcement Learning

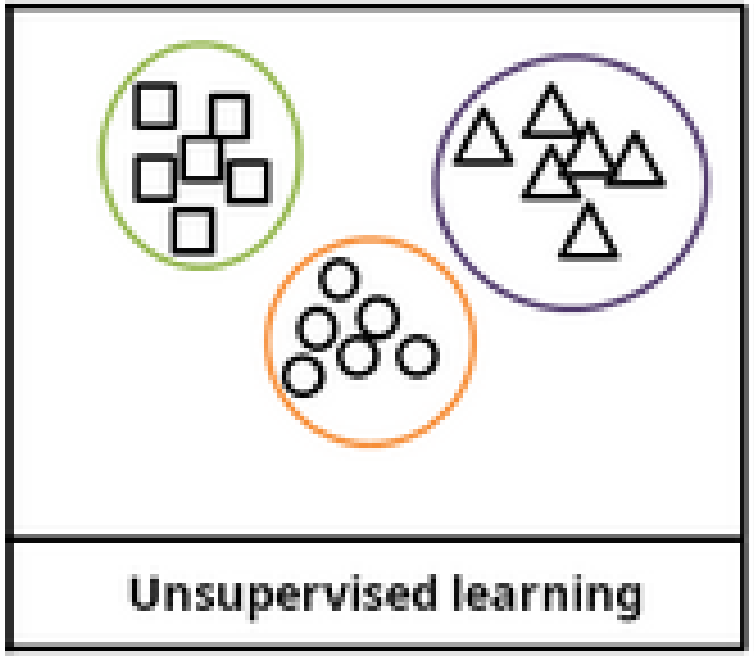




Game Theory applications,
Not as useful in earth sciences

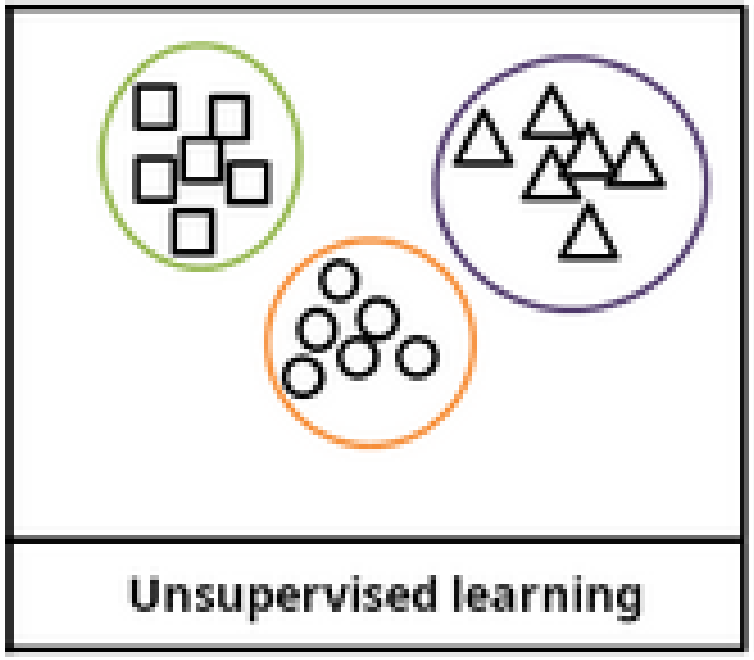
Machine Learning

2. Unsupervised Learning



Unsupervised learning

2. Unsupervised Learning

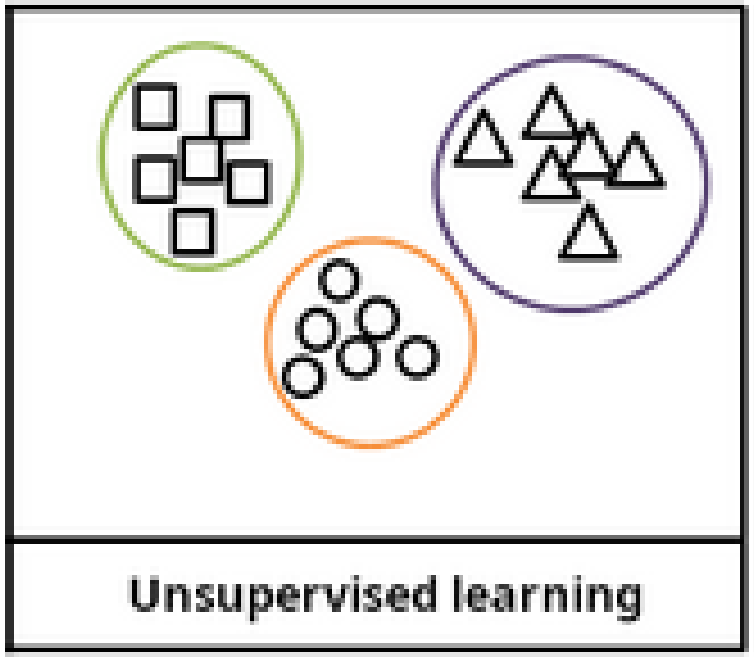


Use it when you want to:

- Reduce the dimensionality of your data
- Find patterns in data without predicting anything per se
- Find a way to group your data that is “creative”

Unsupervised learning

2. Unsupervised Learning

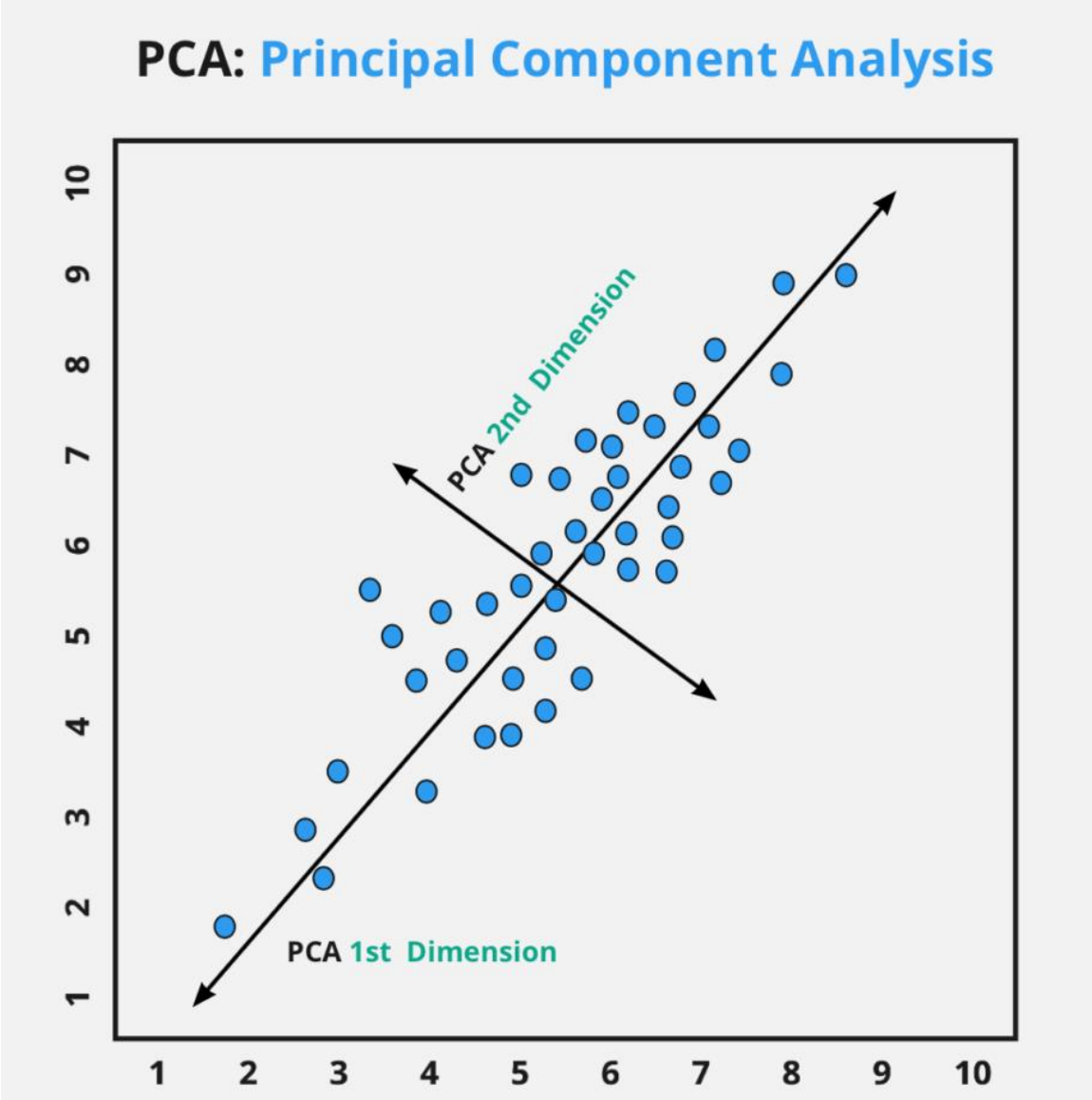


Two examples:

- Principal Component Analysis (PCA) = Empirical Orthogonal Function (EOF)
- K-Means Clustering

Unsupervised learning

- PCA
  - Creates “components” that explain the variance in your data
  - First 1-3 components usually explain most of the variance



Unsupervised learning

- PCA

Environmental Management (2016) 58:1091–1104
DOI 10.1007/s00267-016-0739-6

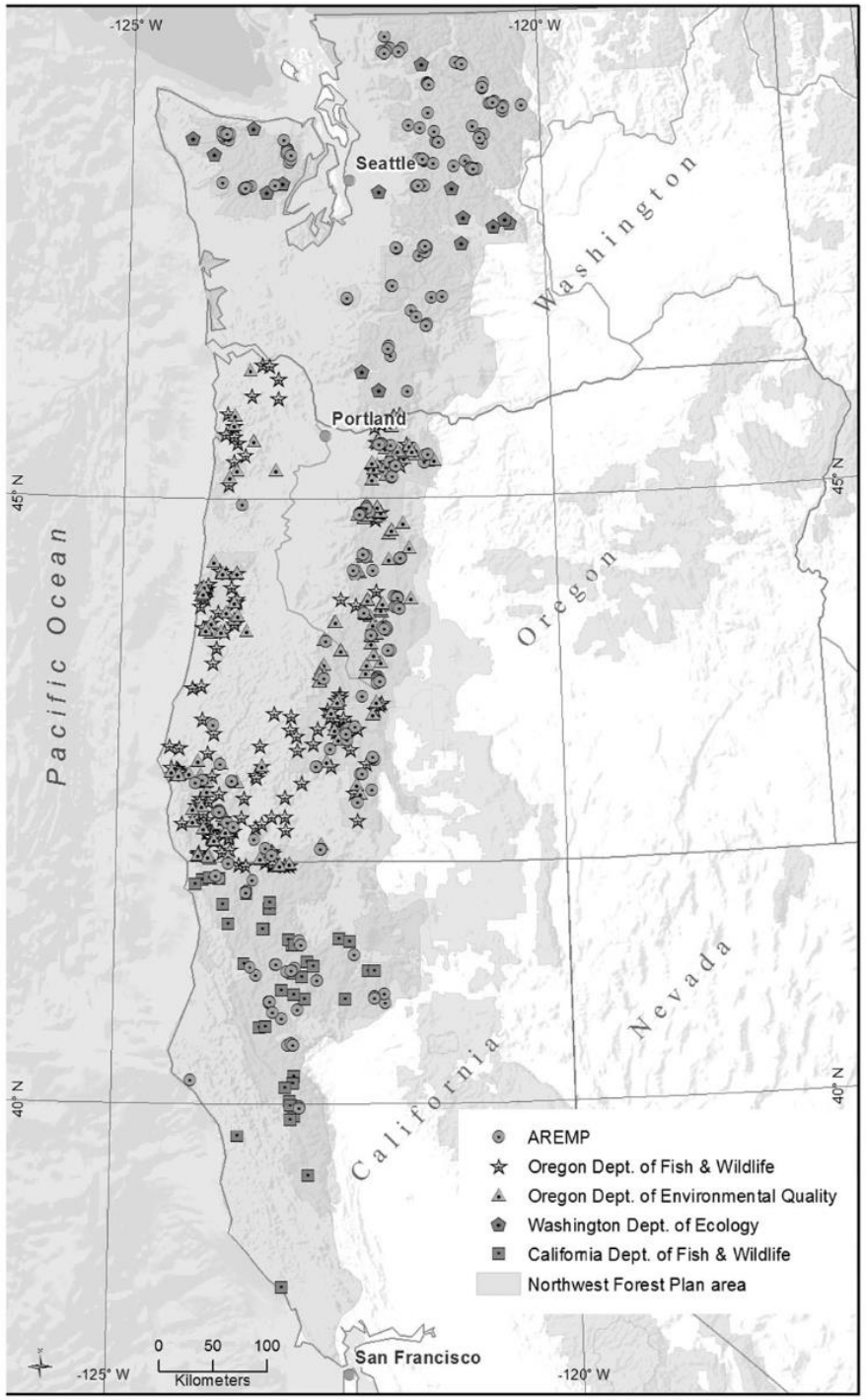
CrossMark

ENVIRONMENTAL ASSESSMENT

A Large-Scale, Multiagency Approach to Defining a Reference Network for Pacific Northwest Streams

Stephanie Miller[1] · Peter Eldred[1] · Ariel Muldoon[2] · Kara Anlauf-Dunn[3] · Charlie Stein[3] · Shannon Hubler[4] · Lesley Merrick[4] · Nick Haxton[4] · Chad Larson[5] · Andrew Rehn[6] · Peter Ode[6] · Jake Vander Laan[7]

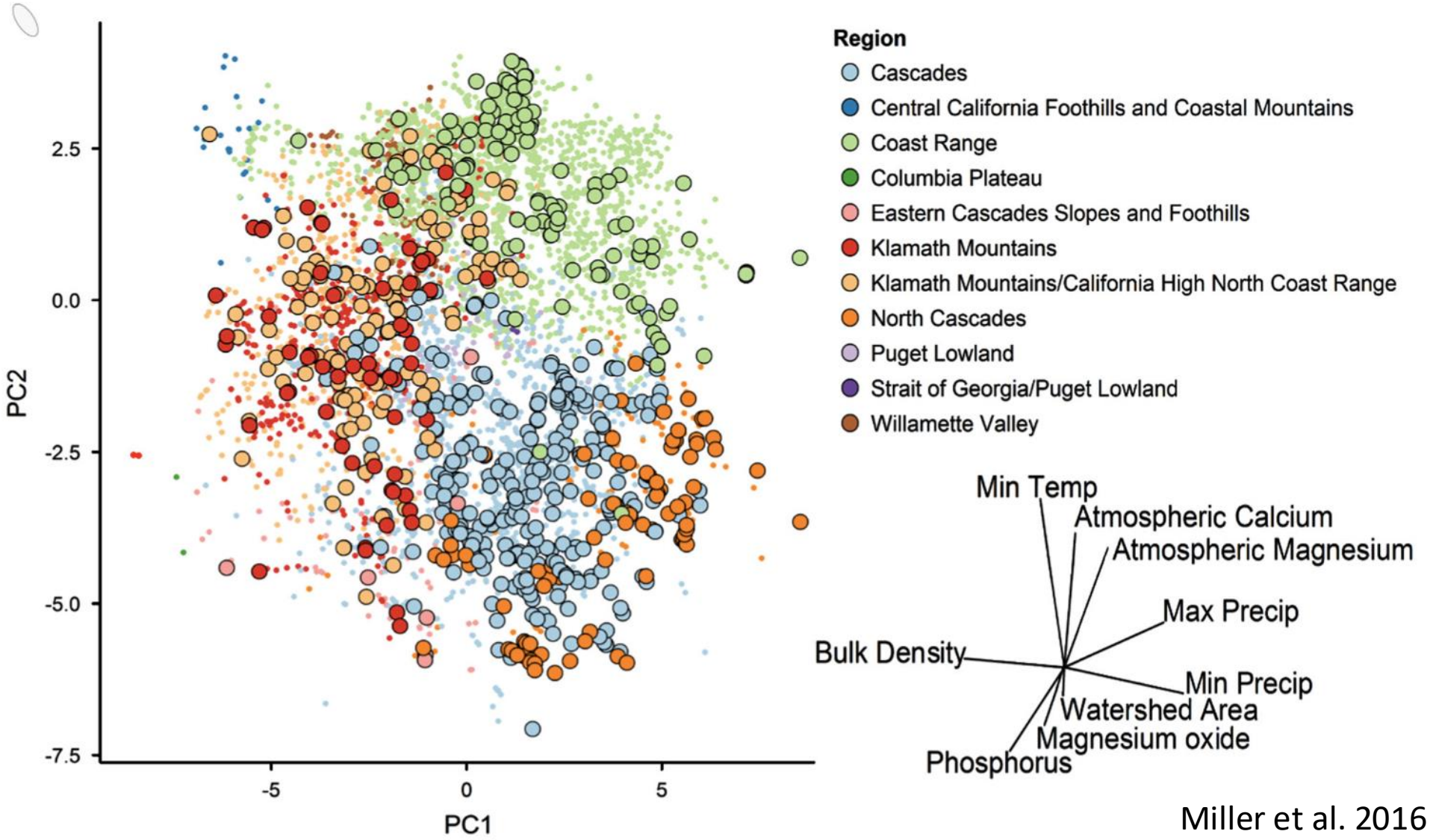
Pacific NW Streams



Unsupervised learning

- PCA

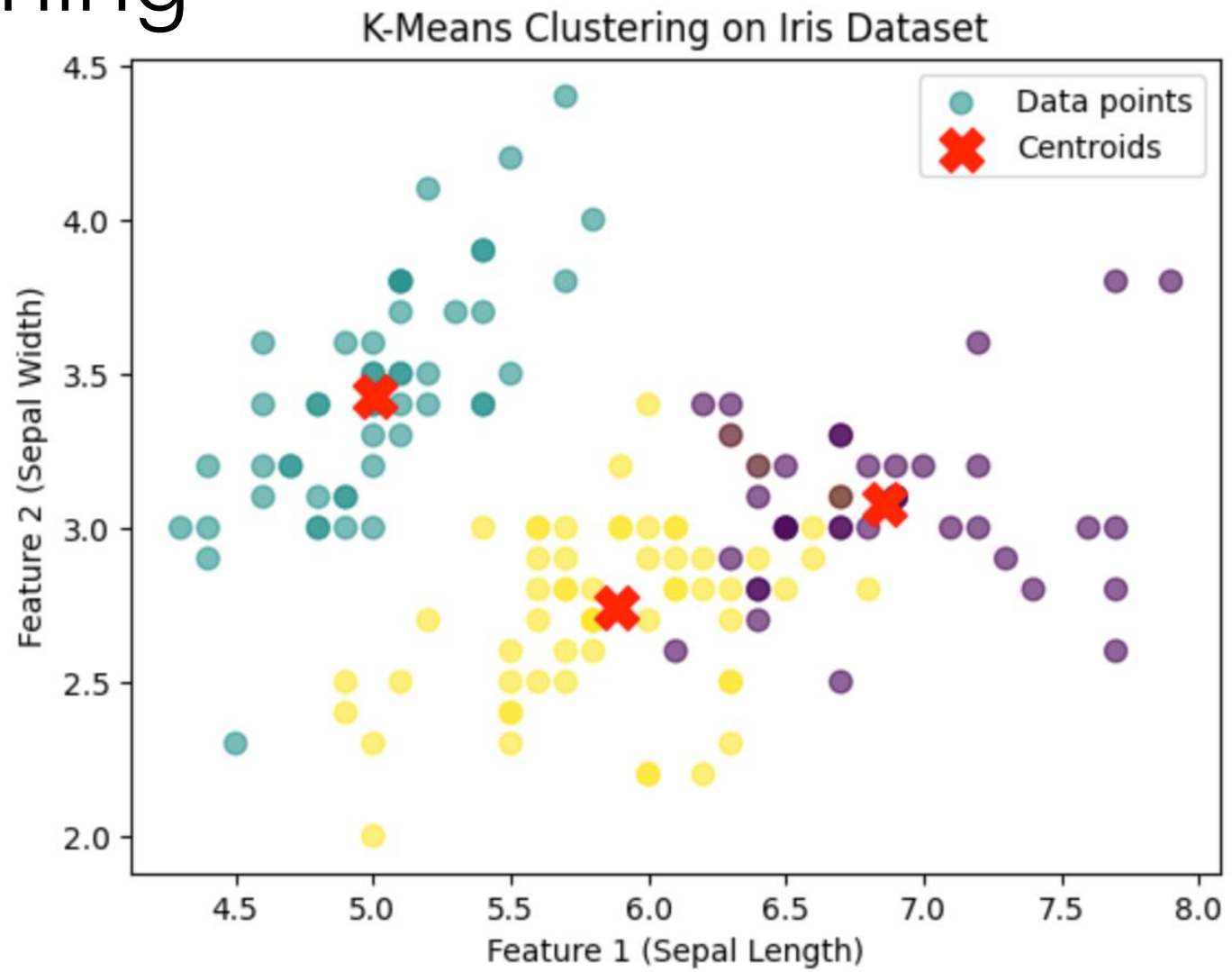
Pacific NW Streams



Miller et al. 2016

Unsupervised learning

- K-means clustering
  - organize messy data into groups based on how similar items are
  - You pick the number of clusters = k



Unsupervised learning

- K-means clustering

remote sensing

MDPI

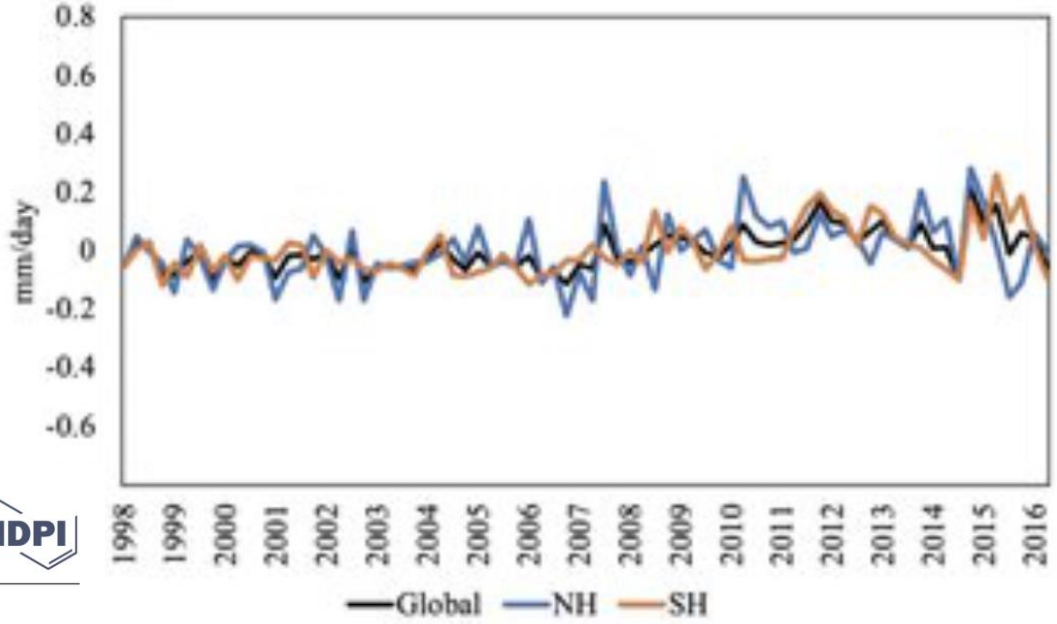
Article

Spatiotemporal Trends and Variations of the Rainfall Amount, Intensity, and Frequency in TRMM Multi-satellite Precipitation Analysis (TMPA) Data

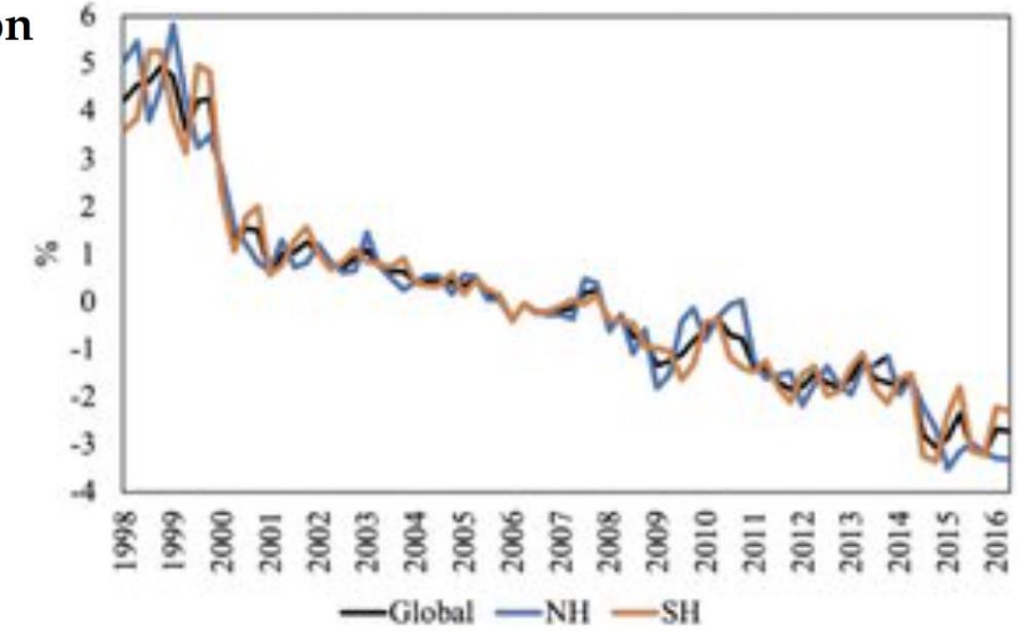
Qian Liu [1,2], Long S. Chiu [3,*], Xianjun Hao [2] and Chaowei Yang [1,2]

Rainfall over the global ocean

Anomalies of Mean Precip. Rate

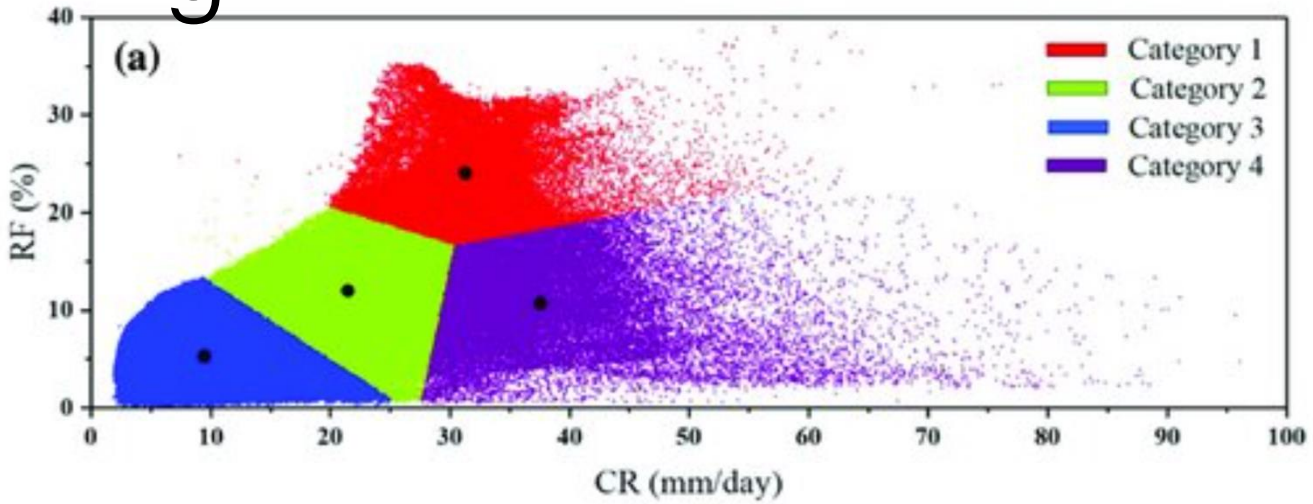


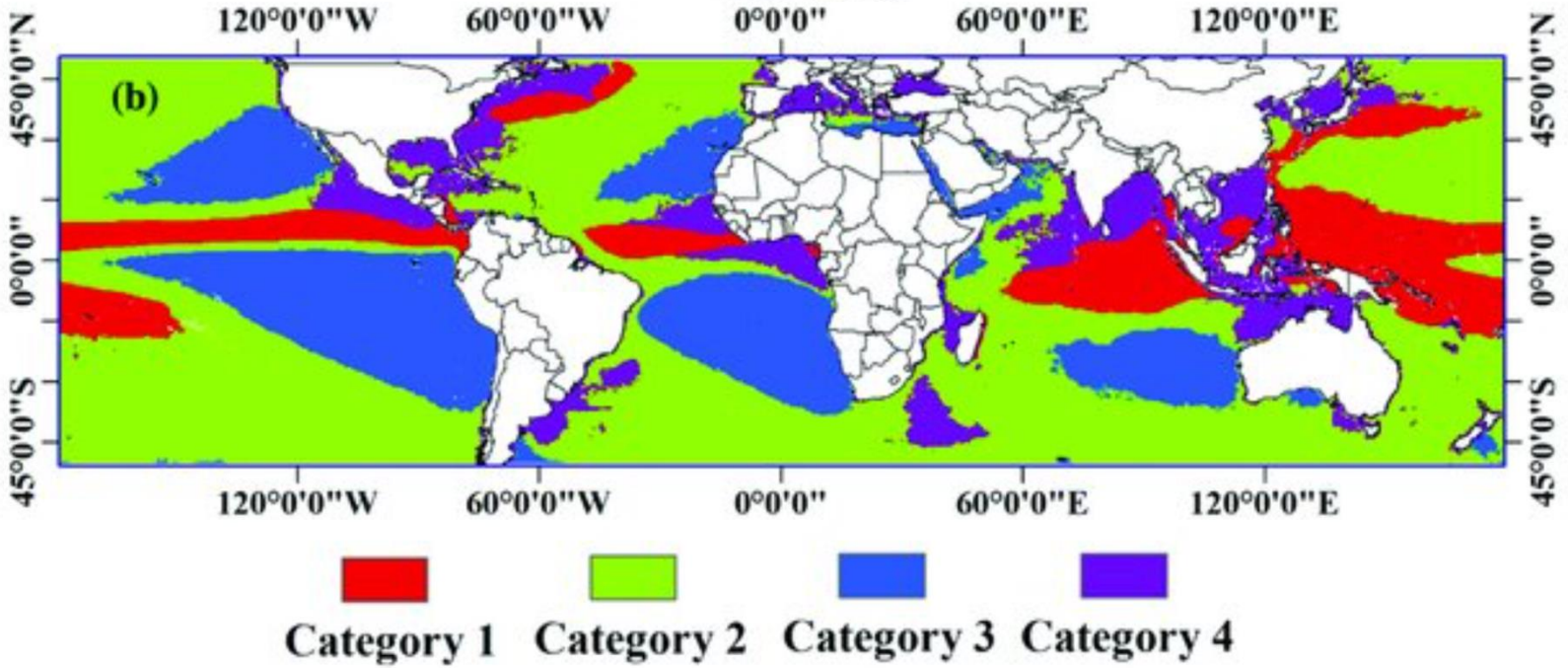
Anomalies of Precip. Frequency



Unsupervised learning

- K-means clustering





Rainfall over the global ocean

Un-supervised ML Models that we will try:

- Principle Component Analysis

PCA_wine_example.ipynb

- K-Means Clustering

Kmeans_unsupervised_example.ipynb

"wine" dataset in scikit-learn



"seeds" dataset from UC Irvine

